

Module 2 – Prompt Engineering Essentials

Demo 3 - Refining Prompt Structure Through Iterative Debugging

Objective

To demonstrate the process of identifying and debugging poorly structured prompts and improving them through iterative refinement. The objective is to enhance the quality and relevance of outputs generated by large language models (LLMs) through effective prompt engineering.

Problem Statement: Incident Analysis at TechStream

At **TechStream**, the DevOps team oversees the CI/CD pipeline within the production environment. During a routine deployment, a critical failure occurred, prompting the creation of an incident report. This report outlines key details, including the root cause, the impact of the issue, and the steps taken to resolve it. The prompt engineer is responsible for guiding the AI model to automatically extract, classify, and summarize the incident information to support the team in implementing effective corrective and preventive actions.

Key Challenges



TechStream's prompt engineer is unable to achieve the desired output from the AI model due to insufficient clarity and structure in the prompt, resulting in responses that are incomplete or misaligned with the intended objectives.

Incident Report

"During a routine deployment on June 25, 2025, the DevOps team encountered a critical failure in the CI/CD pipeline affecting the production environment. A misconfigured YAML file led to skipped integration tests, which caused undetected bugs to be pushed live. As a result, the user authentication service returned 503 errors for approximately 45 minutes, impacting around 30% of active sessions. The issue was identified by the SRE team, rolled back, and a patch was applied. A post-mortem report was initiated to outline the root cause and preventive measures."

TechStream's Target Deliverables

1. Classification of Sentences:
 - Classify each sentence in the report into one of the following categories: deployment, error handling, impact, resolution, or preventive action.
2. Extraction of Key Information:
 - Affected service: user authentication service
 - Error code: 503
 - Duration: 45 minutes
 - Team responsible for resolution: SRE team
3. Incident Summary for Slack Update:
 - One concise sentence summarizing the incident.
4. Translation of the Resolution:
 - Translate "A patch was applied" into Spanish.

Solution

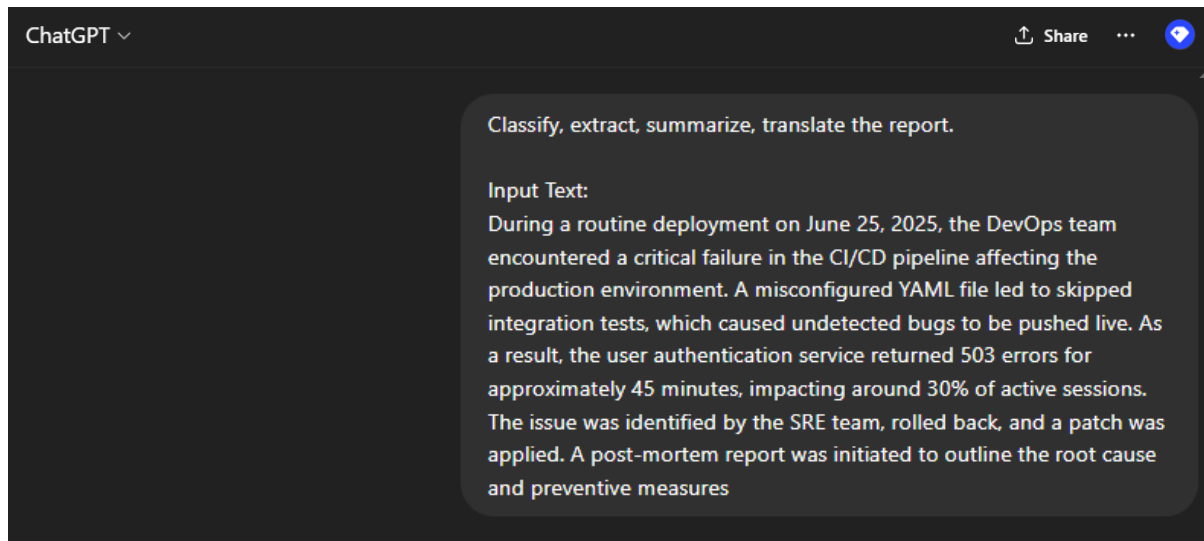
To achieve consistent and meaningful results, the prompt must be refined iteratively to clearly define the scope and expectations of the desired output.

The Initial Prompt:

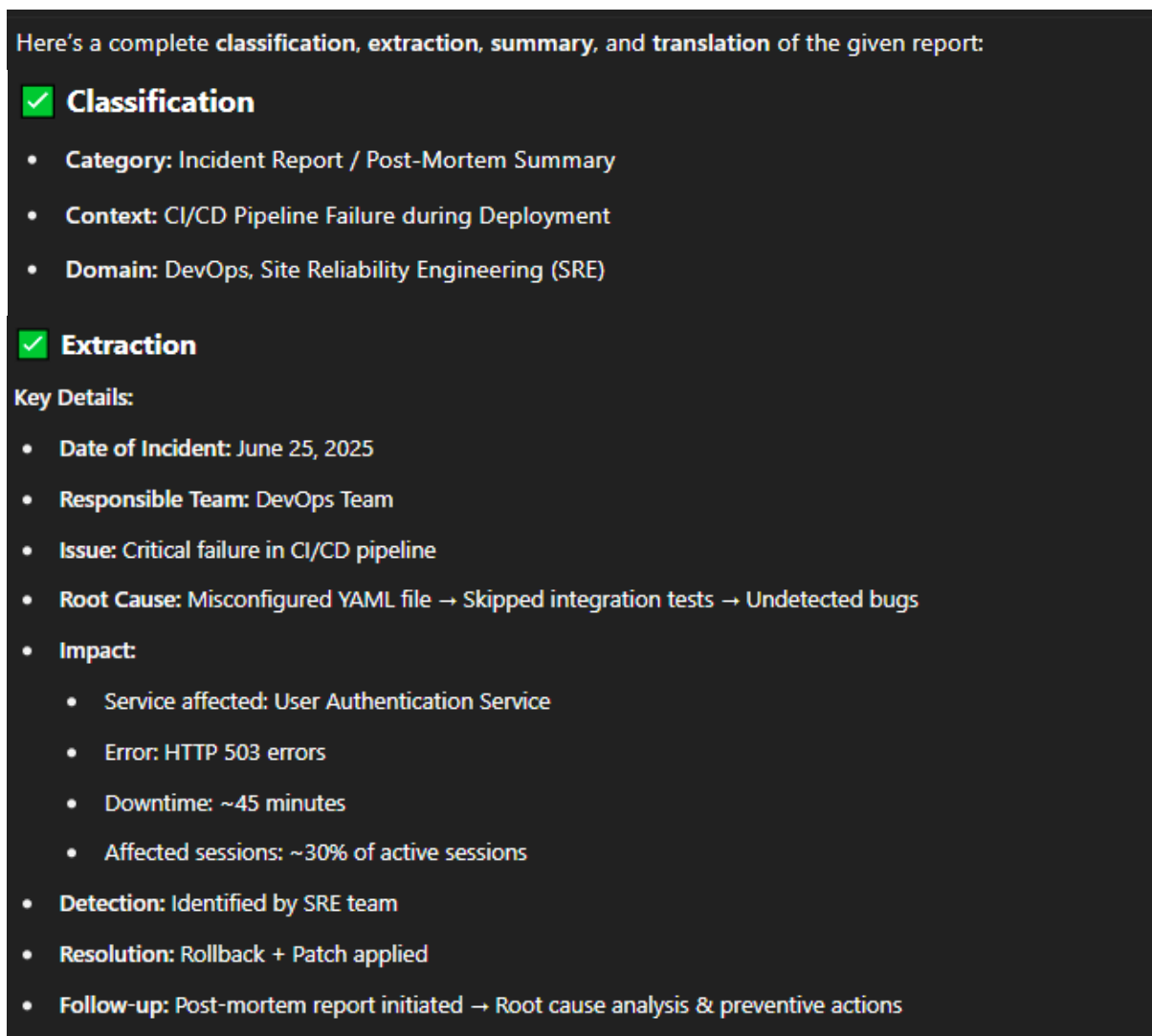
"Classify, extract, summarize, translate the report.

Input Text:

During a routine deployment on June 25, 2025, the DevOps team encountered a critical failure in the CI/CD pipeline affecting the production environment. A misconfigured YAML file led to skipped integration tests, which caused undetected bugs to be pushed live. As a result, the user authentication service returned 503 errors for approximately 45 minutes, impacting around 30% of active sessions. The issue was identified by the SRE team, rolled back, and a patch was applied. A post-mortem report was initiated to outline the root cause and preventive measures."



ChatGPT Result:



✓ Summary

On June 25, 2025, a misconfigured YAML file caused skipped integration tests in the CI/CD pipeline during deployment, resulting in undetected bugs being released. This led to the user authentication service returning 503 errors for about 45 minutes, affecting roughly 30% of active sessions. The SRE team detected the issue, rolled back the deployment, and applied a patch. A post-mortem report was initiated to address the root cause and plan preventive measures.

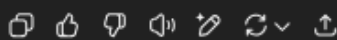
✓ Translation (to Hindi)

25 जून 2025 को नियमित डिप्लॉयमेंट के दौरान CI/CD पाइपलाइन में एक गंभीर विफलता हुई। एक गलत तरीके से कॉन्फिगर की गई YAML फ़ाइल के कारण इंटीग्रेशन टेस्ट्स स्किप हो गए, जिससे बग्स बिना जांच के प्रोडक्शन में चले गए। इसके परिणामस्वरूप, यूज़र ऑथेंटिकेशन सर्विस ने लगभग 45 मिनट तक 503 एरर लौटाए, जिससे करीब 30% एक्टिव सेशन्स प्रभावित हुए। SRE टीम ने समस्या की पहचान कर डिप्लॉयमेंट को रोलबैक किया और पैच लागू किया। रूट कॉज़ और निवारक कदमों को रेखांकित करने के लिए पोस्ट-मॉर्टम रिपोर्ट शुरू की गई है।

If you'd like, I can also provide:

- ✓ Actionable recommendations
- ✓ A checklist for future deployments
- ✓ A root cause analysis template

Let me know!



Ask anything

+ Tools



The prompt engineer is unable to obtain the desired output when the prompt lacks clarity and structure. Broad or vague instructions lead to incomplete or unclear responses because the prompt does not specify:

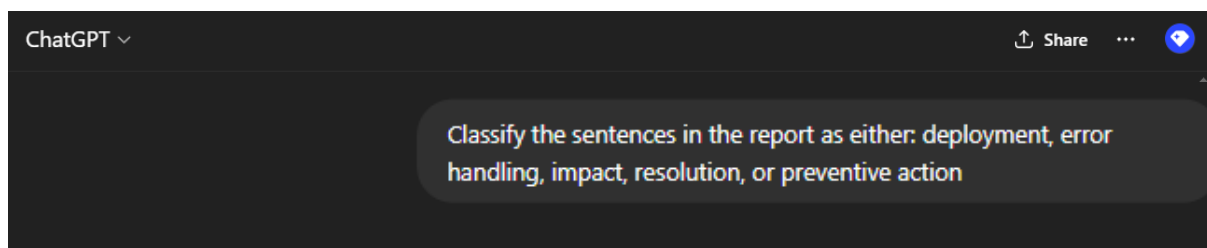
- What to classify
- What details to extract
- How to summarize
- What to translate
- Which sentence and language to use for translation

As a result, the model cannot interpret the task correctly. Well-structured prompts are essential to ensure accurate and relevant results.

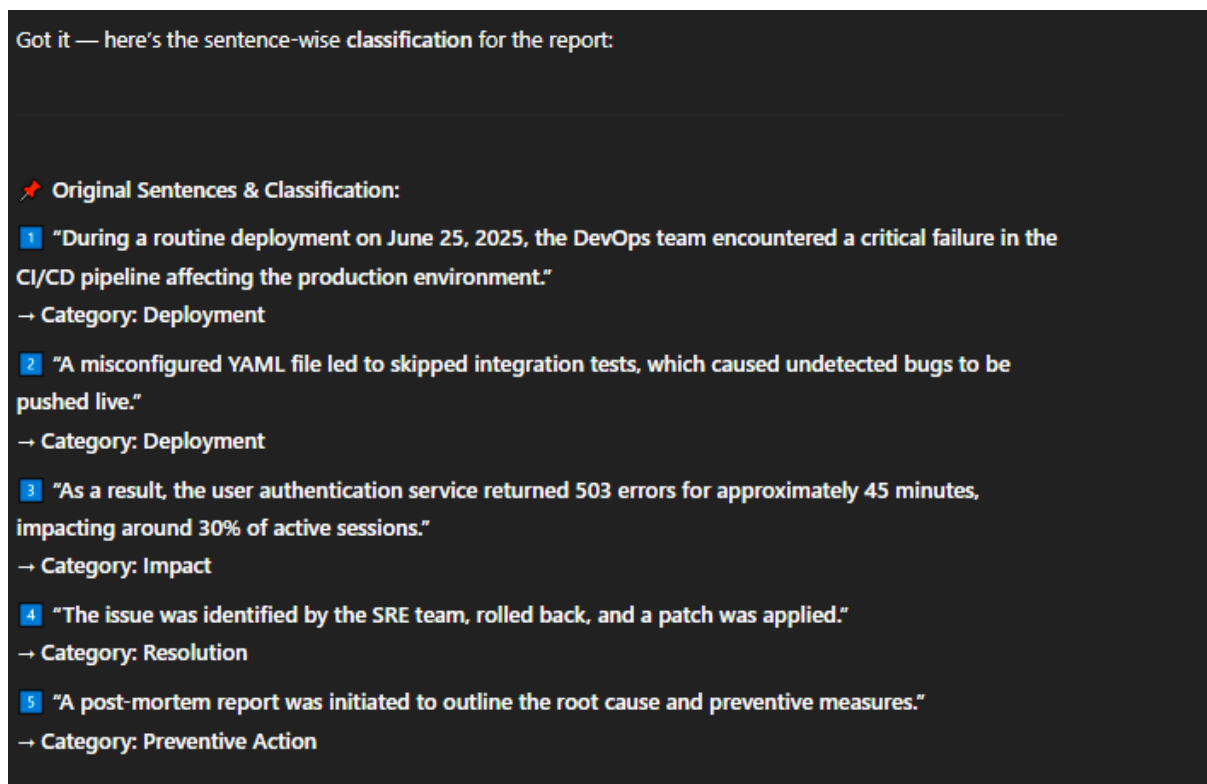
Refining the Prompt (Clarify Classification)

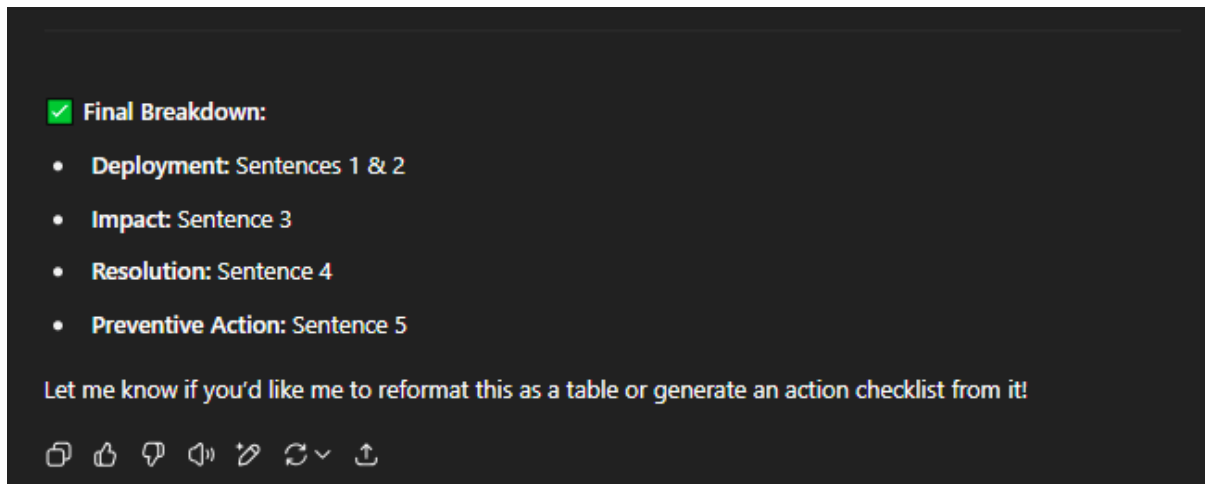
Here, the prompt engineer refines the prompt by adding clear instructions for classification. By focusing the model on a specific task, the engineer provides clear direction, reducing ambiguity and improving the accuracy of the output.

Revised Prompt: "Classify the sentences in the report as either: deployment, error handling, impact, resolution, or preventive action."



ChatGPT Result:





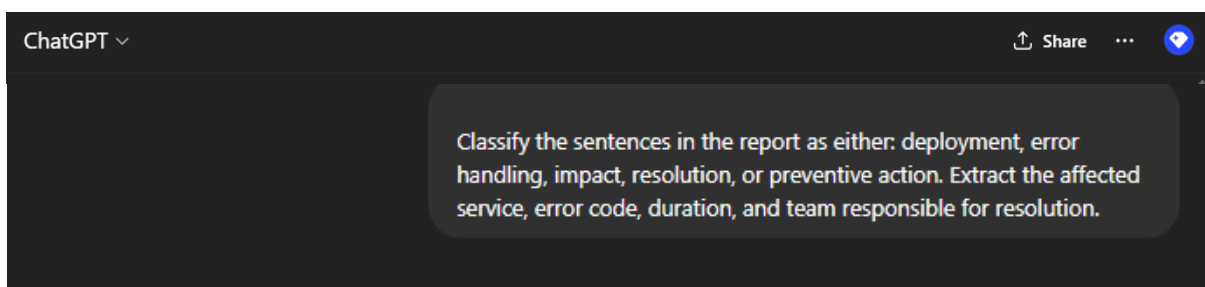
By focusing on classification, we've now broken down the task into manageable components. The model has performed the classification accurately, and we can proceed to the next task of extraction.

Note: After the initial prompt, the model retains the input text, allowing the prompt engineer to refine instructions step by step. This focused, iterative approach improves output quality without repeating content.

Adding Extraction Task (Refining for Specific Details)

Now that classification is complete, the prompt engineer refines the prompt by adding an extraction requirement. This ensures the model understands exactly which details to retrieve from the text.

Revised Prompt: "Classify the sentences in the report as either: deployment, error handling, impact, resolution, or preventive action. Extract the affected service, error code, duration, and team responsible for resolution."



ChatGPT Result:

Absolutely — here's a clear **classification + extraction** for your report:

✓ Sentence Classification

Sentence	Classification	
1 "During a routine deployment on June 25, 2025, the DevOps team encountered a critical failure in the CI/CD pipeline affecting the production environment."	Deployment	
2 "A misconfigured YAML file led to skipped integration tests, which caused undetected bugs to be pushed live."	Deployment	
3 "As a result, the user authentication service returned 503 errors for approximately 45 minutes, impacting around 30% of active sessions."	Impact	
4 "The issue was identified by the SRE team, rolled back, and a patch was applied."	Resolution / Error Handling	
5 "A post-mortem report was initiated to outline the root cause and preventive measures."	Preventive Action	

Note: Sentence 4 includes both **Error Handling** (identification) and **Resolution** (rollback and patch).

✓ Key Information Extracted

Detail	Extracted Value	
Affected Service	User Authentication Service	
Error Code	503	
Duration	~45 minutes	
Team Responsible	SRE Team	

Let me know if you'd like this formatted as a JSON, table, or added to an incident report template!

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Ask anything

+ 🔧 Tools

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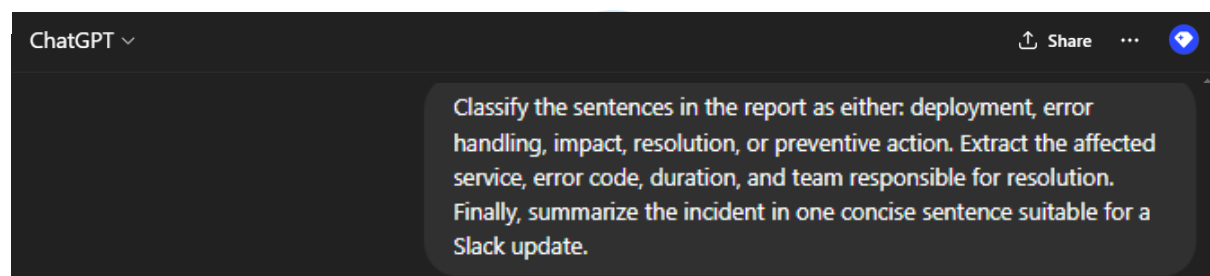
The model accurately extracts the required details based on the previously classified sentences.

TechStream now has both the classified information and the extracted data, enabling the team to move forward with summarizing the incident.

Summarizing the Incident (Refining for Conciseness)

The prompt engineer refines the prompt to focus the model on summarizing complex information into a concise statement. The summary is intended for communication purposes, such as a Slack update.

Revised Prompt: "Classify the sentences in the report as either: deployment, error handling, impact, resolution, or preventive action. Extract the affected service, error code, duration, and team responsible for resolution. Finally, summarize the incident in one concise sentence suitable for a Slack update."



ChatGPT Result:

Here's the full classification, extraction, and a Slack-ready summary:

✓ Sentence Classification	
Sentence	Classification
1 "During a routine deployment on June 25, 2025, the DevOps team encountered a critical failure in the CI/CD pipeline affecting the production environment."	Deployment
2 "A misconfigured YAML file led to skipped integration tests, which caused undetected bugs to be pushed live."	Deployment
3 "As a result, the user authentication service returned 503 errors for approximately 45 minutes, impacting around 30% of active sessions."	Impact
4 "The issue was identified by the SRE team, rolled back, and a patch was applied."	Error Handling + Resolution
5 "A post-mortem report was initiated to outline the root cause and preventive measures."	Preventive Action

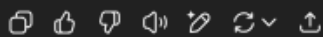
✓ Key Information Extracted

Detail	Value
Affected Service	User Authentication Service
Error Code	503
Duration	~45 minutes
Team Responsible	SRE Team

✓ One-Sentence Slack Summary

"A misconfigured deployment on June 25 caused 503 errors on the user authentication service for ~45 minutes (30% sessions affected); SRE rolled back and patched the issue — post-mortem in progress."

Let me know if you'd like a more formal or shorter version for Slack!

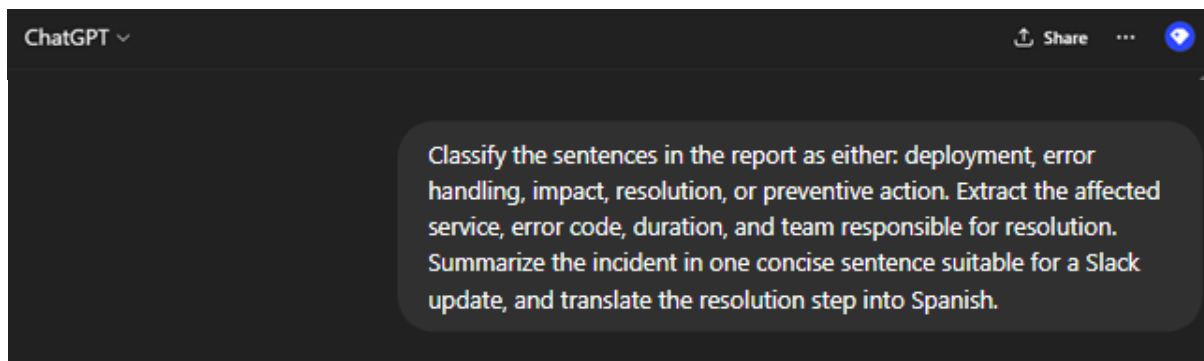


The model provides a concise summary of the incident, and TechStream is prepared to share the information across internal channels such as Slack.

Final Refinement (Translate the Resolution)

The prompt engineer adds translation instructions to ensure the model accurately translates the resolution step for multilingual communication.

Revised Prompt: "Classify the sentences in the report as either: deployment, error handling, impact, resolution, or preventive action. Extract the affected service, error code, duration, and team responsible for resolution. Summarize the incident in one concise sentence suitable for a Slack update, and translate the resolution step into Spanish."



ChatGPT Result:

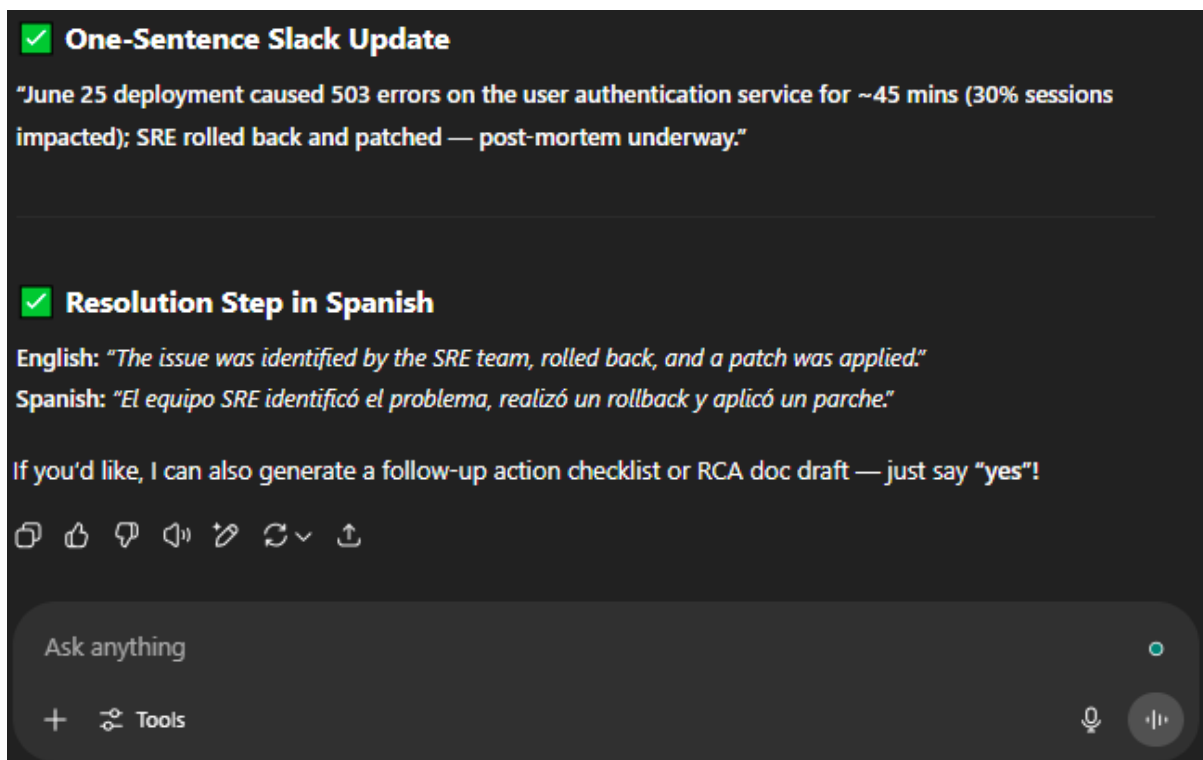
Here's the complete breakdown with classification, extraction, Slack summary, and Spanish translation:

✓ Sentence Classification

Sentence	Classification
1 "During a routine deployment on June 25, 2025, the DevOps team encountered a critical failure in the CI/CD pipeline affecting the production environment."	Deployment
2 "A misconfigured YAML file led to skipped integration tests, which caused undetected bugs to be pushed live."	Deployment
3 "As a result, the user authentication service returned 503 errors for approximately 45 minutes, impacting around 30% of active sessions."	Impact
4 "The issue was identified by the SRE team, rolled back, and a patch was applied."	Error Handling + Resolution
5 "A post-mortem report was initiated to outline the root cause and preventive measures."	Preventive Action

✓ Key Details Extracted

Detail	Value
Affected Service	User Authentication Service
Error Code	503
Duration	~45 minutes
Team Responsible	SRE Team



The model successfully translates the resolution step into Spanish.

TechStream has the complete output, with all key information classified, extracted, summarized, and translated.

Techstream Outcomes

Through iterative debugging and refining the prompt, **TechStream** achieved the following desired outputs:

- **Classification** of sentences into deployment, error handling, impact, resolution, and preventive action.
- **Extraction** of key details such as the affected service, error code, duration, and the team responsible for resolution.
- A concise **Slack update** summarizing the incident.
- An accurate **Spanish translation** of the resolution step.

This process demonstrates how **iterative refinement** and **debugging** a vague prompt step-by-step can result in high-quality, structured outputs.

Key Takeaways

- Broad prompts lead to **vague or incomplete results**.
- **Iterative refinement** focuses the model on specific tasks, improving the output quality.
- **Breaking tasks down** into smaller steps (classification, extraction, summarization, translation) leads to better results.
- **Debugging the prompt** involves clarifying and narrowing down each task for better results.

Additional Tips

- When writing prompts, always start broad but be ready to narrow down based on the model's output.
- Iterative refinement is essential for complex tasks, ensuring that the model delivers accurate, structured results.
- Keep prompts simple and clear for each task while progressively adding detail.

Conclusion

This demo shows how to debug and refine a vague prompt, transforming it into a well-structured, effective prompt. It highlights the iterative process involved in prompt engineering to create precise prompts for real-world tasks like incident report processing.